

HYDROGENATED NBR (HNBR) SYNTHESIS AND LOW-MOLECULAR-WEIGHT PROCESS PATENTS LANDSCAPE

1. ABSTRACT

This report synthesizes patent evidence on hydrogenated nitrile-butadiene rubber (HNBR), with emphasis on routes that control residual unsaturation, molecular weight, and processability while preserving the characteristic oil, heat, and ozone resistance of nitrile rubber-derived elastomers. The patent set spans two major manufacturing philosophies: post-polymerization modification of NBR by hydrogenation, and integrated degradation/hydrogenation strategies that first reduce NBR chain length and then hydrogenate the remaining olefinic sites. Across the landscape, the target is consistently defined as HNBR or liquid HNBR with reduced Mooney viscosity, narrower or deliberately broadened molecular-weight distribution depending on end use, and controlled acrylonitrile content and hydrogenation degree.

The older patents emphasize catalyst-enabled molecular-weight reduction of NBR prior to or during hydrogenation. EP2473281B1 discloses simultaneous metathesis and hydrogenation of nitrile rubber using ruthenium/osmium catalyst systems in hydrogen, aiming to produce low-molecular-weight HNBR with low residual double bond content. US7951875B2 and US20230340237A1 both describe two-step degradation/hydrogenation workflows in chlorobenzene using Zhan-type catalysts, with the latter targeting liquid HNBR having ACN 15-50%, hydrogenation 75-99.5%, Mw 3,000-60,000, and PDI 2.0-8.0. These routes demonstrate that catalyst choice, hydrogen pressure, temperature, and degradation time strongly govern final Mw, PDI, and hydrogenation degree.

In contrast, EP2478021B1 focuses on the precursor NBR synthesis in organic solvent by free-radical solution polymerization with modifiers/RAFT-like control, followed optionally by hydrogenation. Although it is not a direct HNBR hydrogenation patent, it is relevant because precursor microstructure, monomer feed, and molecular-weight control determine downstream hydrogenation behavior and final HNBR properties. US8957161B2 is adjacent rather than direct: it discloses salt-accelerated metathesis of nitrile rubber using Grubbs-type catalysts, which is relevant as a chain-scission tool for preparing lower-MW HNBR precursors. Overall, the evidence supports solution-phase hydrogenation in chlorobenzene or related solvents, ruthenium-based metathesis/hydrogenation catalysts or Zhan catalysts, and careful control of degradation severity to balance fluidity, mechanical strength, and residual unsaturation.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses salt-accelerated metathesis of nitrile rubber using ruthenium/osmium metathesis catalysts, with the metathesis intended to depolymerize unsaturated nitrile rubber before or in connection with HNBR-related processing. The examples show ring-closing metathesis model reactions with Grubbs II catalyst and LiBr or CsBr, demonstrating that added salts can accelerate metathesis conversion. - Salt: LiBr — Example 7.02 - Salt amount: 1 mg LiBr — Example 7.02 - Salt: CsBr — Example 8.02 - Salt amount: 1 mg CsBr — Example 8.02 - Catalyst amount: 10 mg Grubbs II catalyst — Example 7/8 - Solvent: 0.3 ml chlorobenzene — Example 7 - Solvent: 0.2 ml CDCl₃ — Example 7 - Temperature: room temperature — Example 7/8 - Substrate amount: 0.15 ml diethyl diallylmalonate — Example 7 - Time: 2 minutes before substrate addition — Example 7 - Weight ratio salt(s): metathesis catalysts: 0.01:1 to 10000:1 — Claim 1

Relevant Experimental Evidence - Example 7: LiBr addition accelerated ring-closing metathesis relative to the salt-free control. - Example 8: CsBr also accelerated metathesis, though less strongly than LiBr in the reported time course.

Technical Relevance This patent is relevant because metathesis is a chain-scission route for lowering nitrile rubber molecular weight before hydrogenation to HNBR. It provides catalyst/salt concepts that can influence precursor degradation efficiency and thus the final HNBR molecular-weight profile.

Patent 2

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight hydrogenated nitrile rubber prepared by metathesis-based molecular-weight reduction of nitrile rubber followed by hydrogenation. The process uses solution-phase processing in monochlorobenzene (MCB), co-olefin-assisted metathesis, and catalyst loading control to obtain HNBR with reduced Mw and narrower PDI. - Rubber amount: 200 g — Example 1 - Solvent amount: 1133 g MCB — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Degassing cycles: 3 — Example 1 - Rubber amount: 200 g — Example 2 - Solvent amount: 1133 g MCB — Example 2 - Solid content: 15 wt.-% — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Rubber amount: 75 g — Example 3 - Solvent amount: 1175 g MCB — Example 3 - Solid content: 6 wt.-% — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - Rubber amount: 75 g — Example 4 - Solvent amount: 1175 g MCB — Example 4 - Co-olefin: 1-hexene — Example 4 - Co-olefin amount: 150 g 1-hexene — Example 4 - Degassing cycles: 3 with dry N₂ — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - Mn: 85 kg/mol — starting polymer - Mw: 296 kg/mol — starting polymer - PDI: 3.4 — starting substrate feedstock - PDI: 2.0 — typical product - Mw reduction: up to fivefold — Table 2 summary - PDI minimum: 1.90 — Table 2 summary - Mooney viscosity: 73 — Therban® A3407 (comp.) - Mooney viscosity: 43 — Experiment 1 - Mooney viscosity: 28 — Experiment 2 - Mooney viscosity: 3 — Experiment 3 - Mooney viscosity: 31 — Experiment 4

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis was performed on NBR cement at 15 wt.-% solids with 500 psi-class ethylene pressure and 80° C. catalyst conditions. - Example 2: The same metathesis setup was repeated at identical solids and pressure, supporting reproducibility. - Example 3: Lower solids loading (6 wt.-%) was used to assess the effect on molecular-weight reduction. - Example 4: 1-hexene was used as co-olefin, showing that alternative chain-transfer/metathesis partners can be used. - Example 4: Product data showed substantial Mw reduction and PDI narrowing relative to the starting polymer.

Technical Relevance This patent is directly relevant because it establishes a practical route to low-molecular-weight HNBR via metathesis-derived precursor degradation followed by hydrogenation. The disclosed catalyst loading, solvent, temperature, and co-olefin conditions are useful for tuning precursor chain length and final processability.

Patent 3

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed

- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of nitrile rubbers in organic solvents by free-radical solution polymerization of conjugated diene and acrylonitrile monomers in the presence of a modifier, with optional subsequent hydrogenation to HNBR. The emphasis is on controlling molecular weight and polymer microstructure in solution before any hydrogenation step. - Vazo® 88: 284.9 mg — Example 1 - Vazo® 88: 1.17 mmol — Example 1 - Vazo® 88: 0.31 phm — Example 1 - DoPAT: 204 mg — Example 1 - DoPAT: 0.583 mmol — Example 1 - DoPAT: 0.22 phm — Example 1 - Dimethylacetamide: 50 ml — Example 1 - Dimethylacetamide: 51 phm — Example 1 - Acrylonitrile: 43 ml — Example 1 - Acrylonitrile: 653.3 mmol — Example 1 - Acrylonitrile: 37 phm — Example 1 - 1,3-Butadiene: 90 ml — Example 1 - 1,3-Butadiene: 1077 mmol — Example 1 - 1,3-Butadiene: 63 phm — Example 1 - Pressure: 3 bar — Example 1 - Temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Final conversion: 39% — Example 6 - Regulator amount: 2.22 phm — Example 7 - Polymerization time: 9 h — Example 7 - Final conversion: 48% — Example 7 - NBR #7 Mn: 12,000 g/mol — Example 7 - Styrol: 2.79 g — Example 8 - Styrol: 26.8 mmol — Example 8 - Styrol: 50 phm — Example 8 - Temperature: 100 °C — Example 8 - Reaction time: 13 hours — Example 8 - Final conversion: 9.2% — Example 8 - Vazo® 88: 0.15 phm — Examples 18, 20, 23 - Vazo® 88: 128 mg — Examples 18, 20, 23 - Vazo® 88: 0.523 mmol — Examples 18, 20, 23 - DMAc: 5 ml — Examples 18, 20, 23 - Polymerization time: 2.5 and 6 hours — Examples 18, 20, 23 - Post-dosed initiator: 128 mg Vazo® 88 — Examples 18, 20, 23 - Post-dosed solvent: 5 ml DMAc — Examples 18, 20, 23

Relevant Experimental Evidence - Example 1: Baseline solution polymerization of NBR in DMAc with Vazo® 88 and DoPAT under 3 bar butadiene at 100°C. - Example 6: Terpolymerization with hydroxyethyl acrylate was demonstrated, showing compatibility with functional termonomers. - Example 7: Lower-molecular-weight NBR was intentionally prepared by increasing regulator dosage to 2.22 phm. - Example 8: The low-MW NBR macromer was chain-extended with styrene to form an NBR-block-styrene terpolymer. - Examples 17-23: Controlled solution polymerization with optional HDODA/HEA and initiator re-dosing was used to tune conversion and polymer architecture.

Technical Relevance This patent is relevant because HNBR performance depends strongly on the precursor NBR microstructure, molecular weight, and comonomer content established during polymerization. It provides a precursor-synthesis platform that can be paired with later hydrogenation to produce tailored HNBR grades.

Patent 4

- Patent Number: EP2473281B1

- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-pot or simultaneous metathesis/hydrogenation process for nitrile rubber in the presence of hydrogen, using ruthenium or osmium catalyst systems. The method is designed to reduce molecular weight and hydrogenate residual unsaturation in a single operation, yielding low-molecular-weight HNBR with narrower molecular-weight distribution. - Molecular weight (Mw): 20,000 to 250,000 — inventive process summary - Mooney viscosity (ML 1+4 @ 100 deg. C): 1 to 50 — inventive process summary - MWD/PDI: less than 2.5 — inventive process summary - Degree of hydrogenation: 50 to 100% — background HNBR definition - Residual double bond content: 1 to 18% — background HNBR definition - Mooney viscosity (ML 1+4 at 100°C): below 55 — low-MW target discussion - Number average molecular weight (Mn): <200,000 g/mol — low-MW target discussion - Mooney viscosity (ML 1+4 at 100°C): less than 30 — degradation target discussion - Number average molecular weight (Mn): <70,000 g/mol — degradation target discussion - Weight average molecular weight (Mw): 30,000 to 250,000 — cited prior art - Mooney viscosity (ML 1+4 at 100°C): 3 to 50 — cited prior art - PDI: less than 2.5 — cited prior art - Grubbs I catalyst loading: 0.5 phr — cited prior art discussion - Ruthenium content: 307 ppm — cited prior art discussion - Ruthenium content: 61 ppm — cited prior art discussion - Catalyst amount: 65.6 ppm metal content — Example 1 - Catalyst amount: 0.041 phr — Example 1 - Catalyst amount: 0.045 phr — Example 1 - Catalyst amount: 0.060 phr — Example 1 - Co-catalyst: triphenylphosphine — Example 1 - Co-catalyst amount: 1.0 phr — Example 1 - Reaction time: 0, 60, 120, 180, 240 min — Example 1 - RDB: 100% to 0.4% — Example 1 HNBR 1 profile - RDB: 100% to 3.5% — Example 1 HNBR 2 profile - RDB: 100% to 0.7% — Example 1 HNBR 3 profile - Mw: 113371 g/mol — Example 1 HNBR 1 - Mn: 48352 g/mol — Example 1 HNBR 1 - PDI: 2.34 — Example 1 HNBR 1 - Mooney viscosity: 33.3 — Example 1 HNBR 1 - Mw: 133758 g/mol — Example 1 HNBR 2 - Mn: 54595 g/mol — Example 1 HNBR 2 - PDI: 2.45 — Example 1 HNBR 2 - Mooney viscosity: 41.3 — Example 1 HNBR 2 - Mw: 195304 g/mol — Example 1 HNBR 3 - Mn: 64130 g/mol — Example 1 HNBR 3 - PDI: 3.05 — Example 1 HNBR 3 - Mooney viscosity: 69.3 — Example 1 HNBR 3

Relevant Experimental Evidence - Example 1: Three catalyst systems were compared for simultaneous hydrogenation/metathesis, with HNBR 1 showing the fastest RDB reduction. - Example 1: The inventive catalyst gave lower Mooney viscosity and lower Mw than the industrial benchmark HNBR 3. - Example 1: The data show that catalyst identity and co-catalyst choice strongly affect hydrogenation efficiency and molecular-weight reduction.

Technical Relevance This is one of the most directly relevant patents because it explicitly targets simultaneous metathesis and hydrogenation of nitrile rubber to make low-molecular-weight HNBR. It provides a clear catalyst-based strategy for controlling residual unsaturation, Mw, and Mooney viscosity in a single process step.

Patent 5

- Patent Number: US20230340237A1
- Patent Title: Liquid hydrogenated nitrile-butadiene rubber, preparation method thereof and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of liquid hydrogenated nitrile-butadiene rubber by first degrading NBR with a Zhan catalyst in chlorobenzene and then hydrogenating the degraded polymer under hydrogen at elevated pressure and temperature. The process can also be carried out by direct hydrogenation of NBR in organic solvent under inert gas protection, producing LHNBR with high hydrogenation degree and low molecular weight. - NBR amount: 100 g — Example 1 - ACN content: 33% — Example 1 - ML(1+4): 35 at 100° C. — Example 1 - Chlorobenzene: 500 mL — Example 1 - Reactor volume: 1 L stainless steel reactor — Example 1 - Dissolution temperature: 60° C. — Example 1 - Catalyst dosage: 0.03% of NBR — Example 1 - Degradation temperature: 80° C. — Example 1 - Degradation time: 1.5 hours — Example 1 - Hydrogen pressure: 8 MPa — Example 1 - Hydrogenation temperature: 150° C. — Example 1 - Hydrogenation time: 4 hours — Example 1 - Product yield: >98% — Example 1 - Mw: 37,950 — Example 1 - PDI: 2.1 — Example 1 - Hydrogenation degree: 92% — Example 1 - Iodine value: 24 — Example 1 - Tg: -31.2° C. — Example 1 - Tig: -31.7° C. — Example 1 - Teg: -26.9° C. — Example 1 - Catalyst dosage: 0.04% of NBR — Example 2 - Degradation time: 1 hour — Example 2 - Hydrogenation time: 5 hours — Example 2 - Mw: 44,960 — Example 2 - PDI: 2.2 — Example 2 - Hydrogenation degree: 96% — Example 2 - Iodine value: 14 — Example 2 - Tg: -32.5° C. — Example 2 - Catalyst dosage: 0.05% of NBR — Example 3 - Degradation time: 1 hour — Example 3 - Hydrogenation time: 6 hours — Example 3 - Mw: 45,140 — Example 3 - PDI: 2.2 — Example 3 - Hydrogenation degree: 99% — Example 3 - Iodine value: 8 — Example 3 - Tg: -29.8° C. — Example 3 - Catalyst dosage: 0.06% of NBR — Example 4 - Degradation time: 2 hours — Example 4 - Hydrogenation time: 6 hours — Example 4 - Mw: 243.5 million — Example 4 - PDI: 2.4 — Example 4 - Hydrogenation degree: 99% — Example 4 - Iodine value: 8 — Example 4 - Tg: -30.9° C. — Example 4 - NBR amount: 100 g — Example 5 - ACN content: 33% by weight — Example 5 - ML(1+4): 35 at 100° C. — Example 5 - Catalyst dosage: 0.06% of the NBR — Example 5 - Degradation time: 2.5 hours — Example 5 - Hydrogenation time: 6 hours — Example 5 - Mw: 8210 — Example 5 - PDI: 2.7 — Example 5 - Hydrogenation degree: 99% — Example 5 - Iodine value: 8 — Example 5 - Tg: -42.8°

C. — Example 5 - Mw: 15780 — Example 6 starting LNBR - PDI: 3.2 — Example 6 starting LNBR - Catalyst dosage: 0.03% of NBR — Example 6 - Hydrogenation time: 4 hours — Example 6 - Mw: 16250 — Example 6 - PDI: 3.3 — Example 6 - Hydrogenation degree: 99% — Example 6 - Iodine value: 7 — Example 6 - Tg: -38.8°C . — Example 6 - ACN content: 43% — Example 7 - Mw: 32660 — Example 7 starting LNBR - PDI: 4.3 — Example 7 starting LNBR - Catalyst dosage: 0.02% of NBR — Example 7 - Hydrogenation time: 3 hours — Example 7 - Mw: 33420 — Example 7 - PDI: 4.3 — Example 7 - Hydrogenation degree: 91% — Example 7 - Iodine value: 25 — Example 7 - Tg: -30.2°C . — Example 7 - ACN content: 25% — Example 8 starting LNBR - Mw: 32770 — Example 8 starting LNBR - PDI: 5.5 — Example 8 starting LNBR - Catalyst dosage: 0.03% of NBR — Example 8 - Hydrogenation time: 4 hours — Example 8 - Mw: 33,950 — Example 8 - PDI: 5.6 — Example 8 - Hydrogenation degree: 95% — Example 8 - Iodine value: 13 — Example 8 - Tg: -32.9°C . — Example 8 - Mixing temperature: $30-60^{\circ}\text{C}$. — Example 9 - Mixing time: 8-10 min — Example 9 - Vulcanization: 180°C . $\times$ 8 min — Example 9 - Vulcanization: 150°C . $\times$ 4 hours — Example 9 - Shore hardness: 82 — Example 9 - Breaking strength: 7.6 MPa — Example 9 - Elongation at break: 199% — Example 9 - 100% set tensile strength: 3.1 MPa — Example 9 - Shore hardness: 67 — Example 10 - Breaking strength: 4.8 MPa — Example 10 - Elongation at break: 163% — Example 10 - 100% set tensile strength: 3.5 MPa — Example 10

Relevant Experimental Evidence - Example 1: Two-step degradation/hydrogenation in chlorobenzene produced LHNBR with 92% hydrogenation and Mw 37,950. - Example 2: Increasing hydrogenation time gave 96% hydrogenation and Mw 44,960. - Example 3: Higher hydrogenation degree of 99% was achieved with similar Mw and PDI. - Example 5: More severe degradation produced very low Mw (8210) while retaining 99% hydrogenation. - Example 9: LHNBR was compounded and vulcanized to demonstrate practical mechanical properties after curing.

Technical Relevance This patent is highly relevant because it directly targets liquid HNBR with controlled molecular weight, hydrogenation degree, and processing behavior. It provides explicit operating windows for degradation, hydrogenation pressure, temperature, and catalyst dosage that are useful for designing low-MW HNBR synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473281B1 and US20230340237A1 both target low-molecular-weight HNBR, but EP2473281B1 uses simultaneous metathesis/hydrogenation while US20230340237A1 uses sequential degradation then hydrogenation.
- US7951875B2 also uses metathesis-based chain scission before hydrogenation, but emphasizes co-olefin-assisted degradation and reports product Mw around 54 kg/mol and PDI near 2.0.
- US20230340237A1 operates in chlorobenzene with Zhan catalyst at 80°C . degradation and 150°C . hydrogenation under 8 MPa H_2 , whereas US7951875B2 reports MCB solvent, 80°C .

reactor temperature, and ethylene or 1-hexene co-olefins.

- EP2473281B1 reports catalyst loadings in phr and metal ppm, with HNBR 1 showing Mooney viscosity 33.3 and RDB down to 0.4% after 240 min; this is a more integrated one-pot strategy than the two-step routes.
- EP2478021B1 is upstream of the hydrogenation patents: it controls NBR precursor synthesis in solution using Vazo® 88, DoPAT, and DMAc, which can influence downstream hydrogenation behavior and final HNBR properties.
- US8957161B2 is mechanistically adjacent rather than a direct HNBR hydrogenation patent; it shows salt-accelerated metathesis with Grubbs II catalyst and supports the broader chain-degradation toolbox used to prepare lower-MW HNBR precursors.

4. CONCLUSION

The patent evidence shows that successful HNBR synthesis depends on controlling precursor unsaturation, chain length, and hydrogenation severity. The strongest direct routes are simultaneous metathesis/hydrogenation with ruthenium/osmium catalysts (EP2473281B1) and sequential degradation/hydrogenation with Zhan-type catalysts in chlorobenzene (US20230340237A1), both of which can deliver low Mw, low residual double bond content, and acceptable PDI. For practical low-MW HNBR, the evidence supports solution processing in chlorobenzene or monochlorobenzene-class solvents, catalyst dosages in the low phr or sub-percent range, hydrogenation at elevated pressure (8 MPa in the liquid-HNBR patent), and degradation temperatures around 80° C. followed by hydrogenation near 150° C. Where precursor control is needed, solution polymerization of NBR with controlled regulator dosage and monomer feed is important. Overall, the best-supported synthesis strategy is to combine controlled NBR precursor synthesis with either one-pot metathesis/hydrogenation or staged degradation/hydrogenation to reach Mw roughly 8,000-60,000, hydrogenation 91-99%, and PDI around 2-6 for liquid or processable HNBR grades.

5. REFERENCES

- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US20230340237A1 | Liquid hydrogenated nitrile-butadiene rubber, preparation method thereof and ... | Not disclosed | US | 2023 | <https://patents.google.com/patent/US20230340237A1/en>